

MAT9004 Assignment 3

Due: Friday, 18 October 2024, 23:55 PM

Please write up your solutions neatly! In particular, use notation and terminology correctly and include some necessary comments. Partial marks will be given for showing that you know some aspects of the answer, even if your solution is incomplete. The questions are NOT arranged in order of difficulty, so you should attempt every question.

Both handwritten and typed solutions are accepted. Please make sure you convert all your work into **one single pdf file** as other types might not display properly on Moodle.



1. [5 points] Finding location from range measurements. Consider the problem of determining the location (x, y) of a vehicle. Assume that there are some known locations $(a_i, b_i) \in \mathbb{R}^2$ for $i = 1, \dots, m$ (and some $m > 2$). To determine the location, the distances d_i between the vehicle and the known points can be measured, but these are only approximate due to significant errors in the measurement. Hence determining the location is an optimisation problem of finding the position that minimises an error.
 - (a) [2 points] Write down the loss function $\ell(x, y)$ to be minimised, which measures the average squared error. Here the squared error for any point i is the $((\text{distance between } (x, y) \text{ and } (a_i, b_i)) - \text{measurement } d_i)^2$ which is to be averaged overall all $i = 1, \dots, m$.
 - (b) [3 points] What is the gradient of this function $\nabla \ell(x, y)$ (in terms of a_i, b_i and d_i).
Note: This problem is too hard to solve analytically, but to solve it efficiently numerically you need to be able to write down the formulas for the loss function and gradient.
2. [12 points] Consider the bivariate function $f(x, y) = -2x^2 + 2xy - 3y^2 - 5x + 5y + 7$
 - (a) [2 points] Find $\nabla f(x, y)$.
 - (b) [2 points] Find the Hessian matrix $H(x, y)$ for f .
 - (c) [2 points] Find and all stationary points of the function f and classify them as local min., local max. or saddle point.
 - (d) [2 points] Now consider a domain $D = \{(x, y) \in \mathbb{R}^2 \mid y \leq x\} \cap \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 2 - y\}$. Sketch the domain D .
 - (e) [4 points] Find the global maximum and global minimum of f on D (both the point and the objective value).
3. [8 points] A bakery makes 5 kinds of bread. The bread is baked in batches, with the oven fitting 10 loaves at a time. For each part provide not just a number but also a justification of how you calculated the count.
 - (a) [1 point] How many ways are there to create a single batch of 10 loaves?
 - (b) [2 points] The baker explains that there are 3 kinds of savoury bread and 2 kinds of sweet loaves which must be baked in different batches. Now how many different single batches are there?
 - (c) [1 point] The bakery completes 7 batches in a morning, and is considering alternative orders in which to do batches (since order of batches affects when bread is available for customers). If you wanted to consider all possible orders of all possible collection of 7 batches, how many baking schedules (batches and their order) do you have to consider? (No need to calculate the number, just provide the formula)
 - (d) [4 points] To ensure a steady supply of different kinds of bread and to meet the anticipated demand the baker wants:
 1. At most 4 loaves savoury loaves of the same kind in any savoury batch.
 2. At least 4 loaves of each kind of sweet bread in a sweet loaf batch.
 3. The first two and last two batches must be different flavours (sweet vs savoury)
 How many baking schedules for the 7 batches of 10 loaves are there now?

4. [8 points] A delivery van has leave a warehouse and visit 20 customers before returning to the warehouse. For each of the questions below provide only the formula for how to compute the answer, not the actual number:
- [2 points] How many possible schedules are there? Here a schedule defines the time when each customer, is visited when leaving the warehouse at 9am and traveling directly from one customer to the next with no breaks.
 - [3 points] The workload is a little too high to enable a single driver to make all delivery, so the company decides to split it between 2 drivers that do 10 deliveries each. How many possible schedules are there now? Is this a larger or smaller number than for a single driver?
 - [3 points] Due to restrictions by the customers on delivery times, there are 8 customers that must be in the first 5 of either route, and another 3 customers that must be in the last 4 deliveries. How many schedules for 2 drivers making 10 deliveries are there now?
5. [9 points] A factory produces light bulbs, and each light bulb produced has a probability of 0.95 of being non-defective (success) and a probability of 0.05 of being defective (failure). Assume that the production of each light bulb is independent of the others.
- [2 points] What is the probability that exactly 3 out of 5 light bulbs produced are non-defective?
 - [3 points] If the light-bulbs are packed into packs of 6 bulbs, and the factory makes a box of 20 such packs, what is the expected number of packs containing at least one defective bulb in a box?
 - [4 points] The factory has a target of keeping the expected number of defective packs (i.e. packs containing defective bulbs) to at most 2 packs per box of 20. It has a testing procedure that correctly classifies any non-defective bulb with 100% accuracy and defective bulbs with a probability 80%. Would the test help to improve the likelihood of producing packs without defective bulbs? If so, how many bulbs should be tested in each pack, in order to ensure the target is met?
6. [12 points] The lifetime X of the (non-defective) light bulbs in 1000's of hours is modelled by the following probability density function (PDF):

$$f(x) = \begin{cases} \frac{3}{32}x^2 & \text{for } 0 \leq x < 2 \\ \frac{3}{8}e^{1-x/2} & \text{for } x \geq 2, \\ 0 & \text{otherwise } (x < 0) \end{cases}$$

Hint: $\int x e^{ax} dx = \frac{1}{a} e^{ax} (x - 1/a) + C$ for any $a \neq 0$ where C is the integration constant.

- [2 points] What this the probability that a light bulb will last at least two thousand hours?
 - [4 points] To avoid servicing light bulbs too often, enough light bulbs are to be installed in a room to ensure that we can expect at least 6 to be still operating after 2,000 hours. How many light bulbs should be installed in the room? What is the probability that five or less are operating after 2,000 hours?
 - [3 points] What is the expected lifetime of a light bulb?
 - [3 points] If the manufacturer wants to offer a warranty such that only 1 out of a pack of 6 bulbs are expected to fail within the warranty period, what should be the length of the warranty period?
7. [6 points] Let X and Y be two discrete random variables with the following joint probability distribution:

$Y \backslash X$	0	1
0	0.2	0.3
1	0.1	0.4

- [1 point] Which of the two random variables corresponds to flipping an unbiased coin?
- [2 points] Are X and Y independent? Justify your answer.
- [3 points] Calculate the expected value $E[XY]$ and $\text{Var}[XY]$

Total points: 60